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# Fabrication of three-dimensional scaffolds using precision extrusion deposition with an assisted cooling device

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**Abstract**
In the field of biofabrication, tissue engineering and regenerative medicine, there are many methodologies to fabricate a building block (scaffold) which is unique to the target tissue or organ that facilitates cell growth, attachment, proliferation and/or differentiation. Currently, there are many techniques that fabricate three-dimensional scaffolds; however, there are advantages, limitations and specific tissue focuses of each fabrication technique. The focus of this initiative is to utilize an existing technique and expand the library of biomaterials which can be utilized to fabricate three-dimensional scaffolds rather than focusing on a new fabrication technique. An expanded library of biomaterials will enable the precision extrusion deposition (PED) device to construct three-dimensional scaffolds with enhanced biological, chemical and mechanical cues that will benefit tissue generation. Computer-aided motion and extrusion drive the PED to precisely fabricate micro-scaled scaffolds with biologically inspired, porosity, interconnectivity and internal and external architectures. The high printing resolution, precision and controllability of the PED allow for closer mimicry of tissues and organs. The PED expands its library of biopolymers by introducing an assisting cooling (AC) device which increases the working extrusion temperature from 120 to 250 °C. This paper investigates the PED with the integrated AC's capabilities to fabricate three-dimensional scaffolds that support cell growth, attachment and proliferation. Studies carried out in this paper utilized a biopolymer whose melting point is established to be 200 °C. This polymer was selected to illustrate the newly developed device's ability to fabricate three-dimensional scaffolds from a new library of biopolymers. Three-dimensional scaffolds fabricated with the integrated AC device should illustrate structural integrity and ability to support cell attachment and proliferation.

(Some figures in this article are in colour only in the electronic version)

## 1. Introduction

The field of biofabrication brings together the multidisciplinary field of engineering and integrated sciences to fabricate three-dimensional constructs that aid the generation or regeneration of organic tissues and organs [1, 2]. The means by which this can be done is determined by many factors. These factors include, but are not limited to, fabrication technique, available polymers and target organ. Tissue scaffolds are considered to be essential in the generation of tissues. Without a foundation to support and guide cells in a path specific to the target organ, it is nearly unattainable to generate functional tissues [3–8]. Each tissue is unique; some provide protection, while others provide structural framework. The role of each tissue is often determined by the tissue's function, structure, cell configuration and cell type [9–12]. The goal is to leverage geometric positioning and proximity of specific biologics to bring functional abilities to cell aggregates [13–15]. The selection of a viable polymer is often the most difficult task in the fabrication of three-dimensional scaffolds. Nevertheless, the selected polymer must have the ability to be utilized with a fabrication device to construct a scaffold that allows for close mimicry of the target tissue/organ and support cell life [6, 16, 17].

Along with the polymer selection, the device used to create three-dimensional scaffolds often limits the library of polymers available. Thermal extrusion often utilizes polymers whose melting points are typically at least 50 °C. This fabrication technique has the unique ability to create three-dimensional scaffolds of various architectures and mechanical properties [18–21]. In many biofabrication devices, it is often necessary to have a printing resolution on the micron scale. The precision extrusion deposition (PED) device sustains a systematic approach of fabricating three-dimensional scaffolds with the ability to precisely move its motion arms in three-dimensional space [1, 2, 22, 23]. Computer-aided technologies which include, but are not limited to, solid freeform fabrication (SFF), computer-aided manufacturing (CAM) and computer-aided design (CAD) are incorporated to enhance the PED's ability to assemble tissue scaffolds that have similar internal and external architecture as the targeted organ [1, 23–26].

The characteristics and properties of tissue-engineered scaffolds contribute to the success of tissue constructs. Biological models of human tissues illustrate gradients across a spatial volume, in which each identifiable layer has specific functions to perform such that the tissue can function accordingly. These are the characteristics a tissue-engineered scaffold should mimic in order to fulfill the biological and mechanical requirements of the target tissue/organ [27]. Also, the porosity of each tissue-engineered scaffold plays an important role in terms of cell proliferation, attachment and differentiation. The design of the pores within the scaffold should maintain optimal nutrient supply, gas diffusion, active cell proliferation and metabolic waste removal. Literature surveys have demonstrated that large pores allow for effective gas diffusion, nutrient supply and metabolic waste removal while allowing low intracellular signaling and cell attachment. On the other hand, small pores facilitate the exact opposite [28–31].

The big picture in the field of tissue engineering is to first generate a working tissue scaffold that mimics the architecture of the targeted organ that embraces mechanical, chemical and biological cues to sustain cell attachment and proliferation [32–36]. Since cells are seeded onto the scaffolds, the tissue scaffolds are designed to guide the cells to behave similar to that of the target organ. The aided support of growth factors, antibiotics and other supplements enhances the scaffold ability to mature cells into that of the targeted tissue. Once these *in vitro* tissue constructs mature to the stage that is determined to be implantable, the tissue constructs will replace the damaged tissue [37–41]. From the big picture, it is essential that a suitable tissue scaffold be fabricated. Without an optimal tissue scaffold, the task of generating a fully functional tissue organ becomes unattainable.

## 2. Materials and methods

### *2.1. Precision extrusion deposition*

The PED device has three major components: material delivery system, 3D motion and computer-aided modeling. The material delivery component has interchangeable nozzles, two heating elements and a precision drive screw. The heating element heats the material to process temperature, while the precision screw drives the material through the material delivery chamber to the nozzle for extrusion. A schematic view of the material delivery system, along with its temperature gradient, is shown in figure 1. The three-dimensional motion is governed by a set of linear servo motors that permits the PED to move in the $\pm X$, $\pm Y$ and $\pm Z$ directions. The computer-aided modeling component of the PED enables its user to model three-dimensional structures accordingly [1, 2].

As shown in figure 1, there are two heating elements that control the temperature along the material delivery chamber. The top heating element (heating element 1), located at the inlet of the chamber, is typically set at a higher temperature in comparison to the bottom heating element (heating element 2) located at the nozzle [1, 2]. The primary goal of heating element 1 is to heat the biopolymer until it changes the polymer's phase from solid to liquid. The second heating element then maintains the biopolymer's viscosity through the chamber until it is extruded. Heating element 1 has a higher temperature in comparison to heating element 2 because it takes more energy to change the biopolymer's phase from solid to liquid than to maintain the liquid phase. The first heating element is not only responsible for phase changes, it is also responsible for setting the appropriate viscosity that allows for the extrusion of cylindrical filaments.

The velocity, acceleration and deceleration of the $X$, $Y$ and $Z$ axes are independent of each other. However, the extrusion speed (sometimes referred to as the fourth axis) of the material delivery system is proportional to the velocity of the motion. Depending on the viscosity of the working material, the proportional gain must be adjusted accordingly to produce a smooth flow (no overflow or underflow of materials) of filaments as the PED builds the three-dimensional structures. The proportional gain is the speed of the extrusion screw; the speed is proportional to the velocity of the $X$, $Y$ and $Z$ linear slides. The working temperature on the material delivery chamber is optimized to maintain the proper viscosity for extrusion. The appropriate viscosity for extrusion is determined by the material's mechanical properties and the micro nozzle diameter. The motion velocities of the $X$, $Y$ and $Z$ linear slides are also optimized to ensure

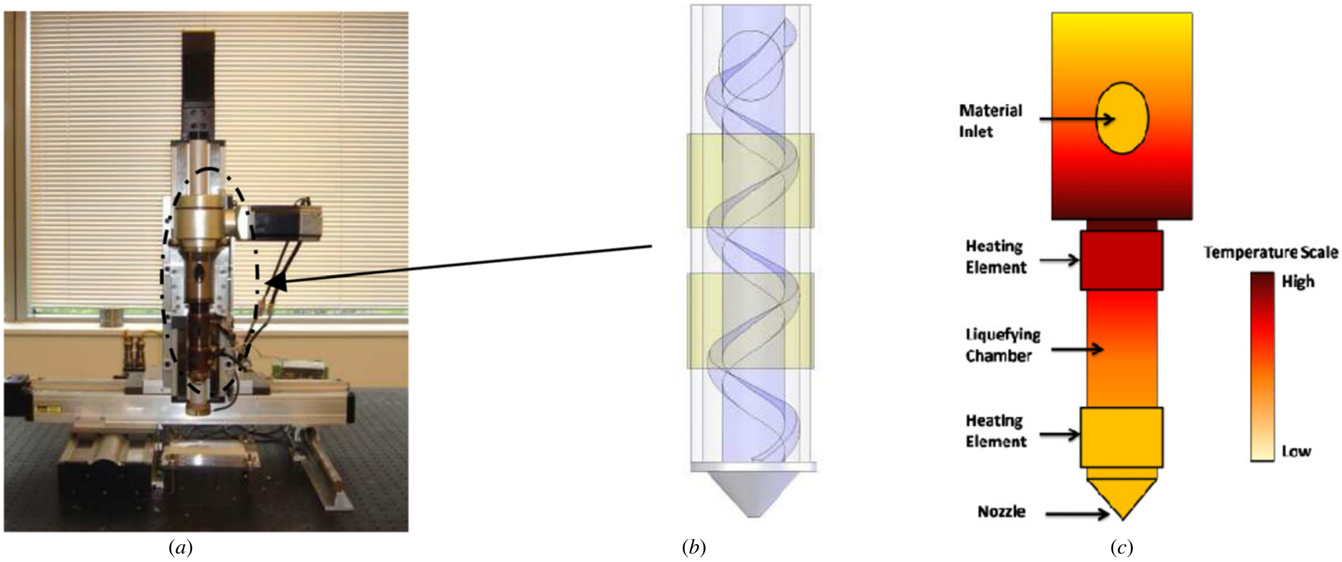



**Figure 1.** (*a*) Front view of the PED, (*b*) schematic view of the material delivery chamber, (*c*) temperature gradient schematic of the material delivery chamber where the top heat element has a higher working temperature in comparison to the lower heating element.

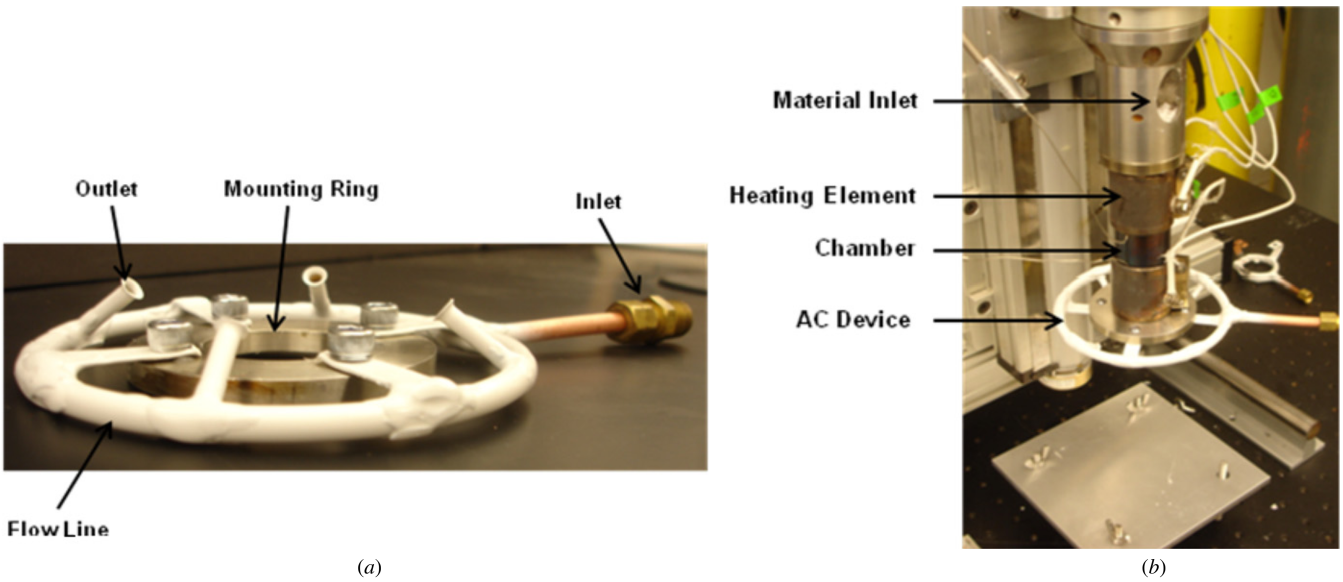



**Figure 2.** (*a*) AC device, (*b*) AC device integrated on the material delivery chamber of the PED device.

that there are no stretching or clumping of the extruded filament. The appropriate velocities eliminate the possibilities of deformation.

In terms of the material delivery system, the two heating elements are regulated at their respective desired working temperature, and then the polymer is inserted at the inlet (as shown in figure 1(*c*), the inlet is located above the top heating element). Polymers need to be of the form of small pellets no greater than 15 $ml^{-3}$ for extrusion. Once inserted, the rotational screw (computer controlled) within the material delivery chamber transports the polymer pellet toward the nozzle for extrusion. While the polymer is moving through the chamber, the heat would melt the polymer to its appropriate viscosity. Pellets are continuously fed until the fabrication process is completed.

### *2.2. Assisting cooling device*

The PED by itself cannot extrude high melting point biopolymers. The demand for a device that would fabricate scaffolds from high melting point biopolymers has increased significantly. These demands require a device that has the ability to be integrated onto the PED and cool the extruded filaments as they are extruded. This device should not manipulate the material delivery chamber's temperature, induce rapid cooling or deform the extruded filament's geometry in any shape or form. The assisting cooling (AC) device uses, but is not limited to, nitrogen (gaseous), air, oxygen, chilled water and culture medium to cool the filaments as they are extruded from the nozzle via convective and conductive cooling. As shown in figure 2, the AC device will be mounted near the nozzle of the PED. Heat from the

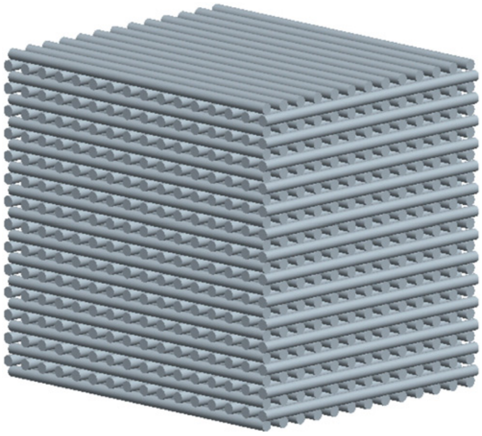


**Figure 3.** Modeled uniform 0°/90° filament orientation scaffold.

material delivery chamber of the PED has no influence on the AC working medium. The AC has four cooling points located on each quadrant on the flow line; this allows for cooling in each direction of motion on the *XY* plane. The integration of the AC device on the material delivery chamber allows for complete three-dimensional cooling as scaffolds are fabricated.

Prior to the installation of the AC device, the PED processed polymers melting points were up to 100 °C. If the polymers' melting point were above 100 °C, the applied heat would be elevated to facilitate the proper viscosity for extrusion. The issue with extruding polymers with a melting point higher than 100 °C with the PED technology is that once the fabrication process is initiated and the scaffold is being fabricated, the extruded filaments begin to deform (change from the cylindrical structure to a flattened filament). The deformation is due to the fact that the extruded filaments are still hot and are in a liquid-like state. The integration of the AC device addresses this issue. As the polymer is being extruded, a low flow medium dissipates heat from the extruded filament. This allows the filament to retain its cylindricity as the scaffold is being fabricated. Since the AC device cools the polymers as it is being extruded, the range of biomaterial the PED with the integrated AC device can process is increased to polymers with melting points of up to 250 °C. The limitation of 250 °C is enforced by the working conditions of the material delivery servo drives.

Since the cooling media are held constant, it is the flow rate of the cooling medium that has the impact upon the extruded polymer. If the rate is high, the polymer will rapidly be cooling causing the filaments to break as they are patterned into a scaffold. If the rate is low, then the polymers will not dissipate a sufficient amount of heat and the filament will deform (flat) and will create a porous scaffold. The optimization of the cooling medium's flow rate will reduce filament deformation and allow for the fabrication of porous three-dimensional scaffolds.

### *2.3. Scaffold fabrication, morphology and sterilization*

A fairly simple mechanical geometry of a uniform 0°/90° filament orientation scaffold is fabricated to characterize the device's fabrication capabilities. CAD/CAM applications were utilized to create a database of fabrication protocol/ procedures which the PED utilizes to produce scaffolds. Process parameters such as the heating element temperature, motion velocity and extrusion speed are set to ideal settings. At the state of static equilibrium, the database is loaded onto the PED for scaffold fabrication. The modeled uniform 0°/90° filament orientation scaffold is illustrated in figure 3.

To accurately extrude filaments with zero deformation to its structure, the first step is to find the process temperature at which the polymer can be extruded using the PED with the integrated AC. According to the parameters of the PED, there are two temperature parameters that exist for the extrusion of polymer. Through experimental analysis (figure 4), the optimal temperature range for heating element 1 is 200–220 °C, while the optimal temperature range for heating element 2 is 180–200 °C. The rectangular box illustrated in figure 4 represents the optimal working temperature for the proprietary polymer. As seen in figure 4, the vertical axis demonstrates the phase change of the polymer with respect to the temperature change. The appropriate viscosity for extrusion is listed as a liquid phase. The liquid phase is best used for the extrusion of cylindrical filaments. The viscosity of the polymer in the gel and solid phases are too high for extrusion [2, 42].

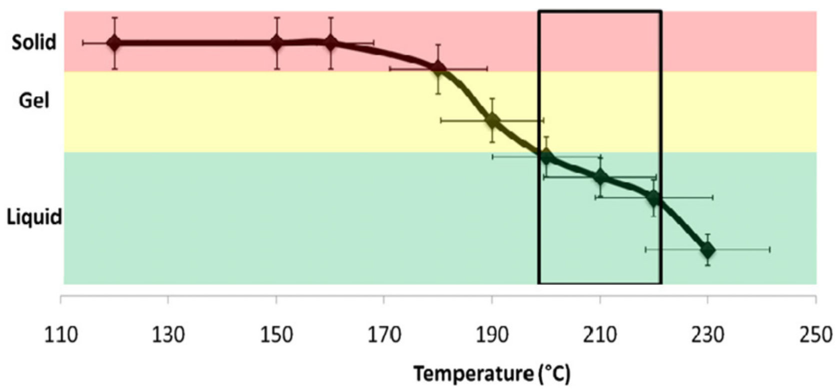



**Figure 4.** Experimental analysis of extrusion temperature, and the successively process window for the extrusion range is shown within the rectangle.

**Table 1.** Properties of modeled three-dimensional uniform porosity scaffolds.

| | Scaffold 1 | Scaffold 2 | Scaffold 2 |
|---|---|---|---|
| Filament orientation | $0^\circ/90^\circ$ | $0^\circ/90^\circ$ | $0^\circ/90^\circ$ |
| Filament size | 350 $\mu$m | 350 $\mu$m | 350 $\mu$m |
| Pore size | 200 $\mu$m | 350 $\mu$m | 500 $\mu$m |
| Dimension ($L \times W \times H$) | ($10 \times 10 \times 5$) mm | ($10 \times 10 \times 5$) mm | ($10 \times 10 \times 5$) mm |
| Porosity | 51.53% | 62.30% | 67.69% |

The slow flow rate of the polymer through the material chamber requires a slow motion velocity for the $X$, $Y$ and $Z$ axes. To have a consistent flow of filaments for scaffold fabrication, the process parameter for the motion velocity ranges from 1.00 to 12.00 mm s$^{-1}$, while the process parameter for the extrusion gain (gain is proportional to the motion velocity) ranges from 1.00 to 2.50. The polymer used in this paper is a proprietary polyglycolide (PGA) polymer (average molecular weight ($M_w$) $\sim$ 242 kDa, melting point 180 °C). Some commercially available polymers that can be utilized on the PED with integrated assisted cooling are polyglycolide (PGA), poly (lactic acid) (PLA), polycaprolactone (PCL), hydroxyapatite (HA) and Polystyrene. Each polymer listed is unique in terms of their mechanical properties and biodegradability. Depending on the target organ/tissue, one or more of these polymers may be utilized to fabricate porous three-dimensional scaffolds.

The process parameters utilized for the fabrication of the scaffolds listed in this paper are as follows: (1) heating element 1: 220 °C, (2) heating element 2: 200 °C, (3) AC medium: nitrogen, (4) AC flow rate: 0.03 kg s$^{-1}$, (5) applied cooling temperature of nitrogen from the outlet of AC device: 16 °C, (6) proportional gain of extrusion screw: 1.25 and (7) motion velocity of linear arms: 2 mm s$^{-1}$.

The porosity of the scaffold is calculated using equation (1), where $V_T$ is the total volume of the scaffold and $V_P$ is the volume of the polymer within the scaffold. The volume of the polymer within the scaffold is calculated using equation (2), where $L$ is the number of layers on the scaffold, $V_f$, is the volume of the $i$th filament. Similar methods of calculating the porosity in relation to the material within the scaffold are discussed by Too *et al* and Shor *et al* [2, 43]:

$$V_A = V_T - V_P \tag{1}$$

$$V_P = L \sum_{i=1}^{n} V_f^i. \tag{2}$$

Experiments conducted in this paper utilized three scaffolds with varying porosity. The properties of these scaffolds are listed in table 1. The various porosities were utilized because of the osteoblast cell line (7F2) used in the biological study. The 7F2 cell line prefers porous scaffolds within the range listed in table 1 [28].

The micro-structural form and internal morphology of the refined fabricated scaffolds were evaluated using an FEI/Philips XL-30 field emission environmental scanning electron microscope (SEM). The images were taken using a beam intensity of 5 and 10 KV and gaseous secondary electron detectors of 1.3 Torr. Before scaffolds were characterized for architectural integrity, appropriate preparation was conducted by first freezing in liquid nitrogen for approximately 5 min, then scaffolds were removed and immediately sectioned using a sharp straight razor. Sectioned scaffolds allowed for the SEM characterization of the internal architecture. The visualization of the internal architecture illustrates the interconnectivity, filament formation, porosity and structural integrity of the fabricated scaffolds. SEM images of each scaffold type (scaffolds 1, 2 and 3) were taken.

All scaffolds were subjected to sterilization prior to the biological study. According to the sterilization protocol, scaffolds were submerged within 95% ethanol and placed on a magnetic stirrer for 24 h. Scaffolds were removed from the ethanol and placed under a sterile environment for drying (overnight) and later stored under sterile conditions.

### 2.4. *Biological exploration*

By definition, biopolymers are biocompatible in every shape and form; however, the applied heat and frictional heat induced onto the polymer during the fabrication process may change the cytotoxicity level within the scaffold. Along with the cytotoxicity of the scaffold, the interactions of the cells within the scaffolds are of interest. Cells seeded onto the scaffold must have the ability to attach, proliferate and differentiate into mature cells. Biological assays such as fluorometric indicators and enzyme-linked immunosorbent assay (ELISA) will be utilized to verify the feasibility of utilizing these polymers as potential building blocks to construct functional tissues.

7F2 (osteoblast) cell line (source: ATCC, Manassas, VA, USA) was seeded onto 75 cm$^2$ vented flasks and incubated. Six hours after the cells were seeded, the culture medium was changed to remove any dead cells in the flask; the culture medium was also changed every 2 to 3 days until flasks were confluent. Confluent flasks were harvested and counted using a hemocytometer. Cells were then centrifuged again in which the cell pallet was suspended to a cell density of $1 \times 10^6$ cells ml$^{-1}$.

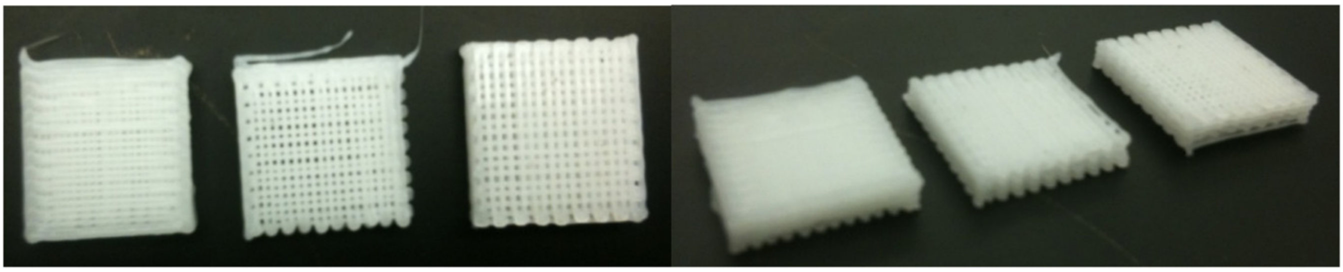
A sterile scaffold was submerged into cell culture media and placed in the incubator for 24 h prior to the cell seeding process. The scaffold was then removed from the culture media and washed once with 1× phosphate buffered saline (PBS). Then, 1 ml of cell suspension ($1 \times 10^6$ cells) was gently pipetted to the scaffold throughout each pore of the scaffold. The scaffold was incubated for 4 h. This period allowed the cells to properly attach to the scaffold. After 4 h, scaffold was moved to a new well plate, and fresh media were added to each well. The scaffold was washed with 1× PBS every 2 to 3 days and fresh media were added.

All biological investigation data are expressed as the mean ± standard deviation for the sample size of 3 ($n = 3$).

## 3. Results and discussions

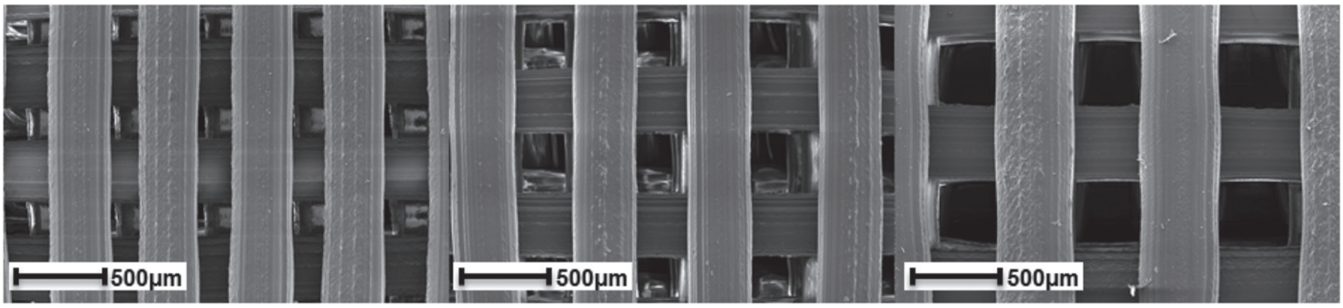
### 3.1. *Scaffold characterizations*

Still images in figures 5(*a*) and (*b*) illustrate the top and isometric views, respectively, of a batch of fabricated scaffolds.



(*a*) (*b*)

**Figure 5.** (*a*) Top view, (*b*) isometric view of fabricated scaffolds (from left to right: scaffold 1, scaffold 2 and scaffold 3).




(*a*) (*b*) (*c*)

**Figure 6.** SEM image showing the 0°/90° filament orientation of (*a*) scaffold 1, (*b*) scaffold 2, (*c*) scaffold 3, magnification: 50×, scale bar: 500 $\mu$m.

These figures reveal the dimension and varying porosity of each scaffold. As seen in figure 5, scaffold 1 is on the left, scaffold 2 is centered and scaffold 3 is on the right.

The scanning electron microscopy (SEM) images deliver evidence that at high extrusion temperature the PED with the integrated AC device is fully capable of fabricating scaffold at the micro-scale level. The SEM images undoubtedly confirm the 0°/90° filament orientation, interconnectivity, structural integrity and porosity of the above-mentioned three-dimensional scaffolds 1, 2 and 3 (figures 6, 7, and table 1).

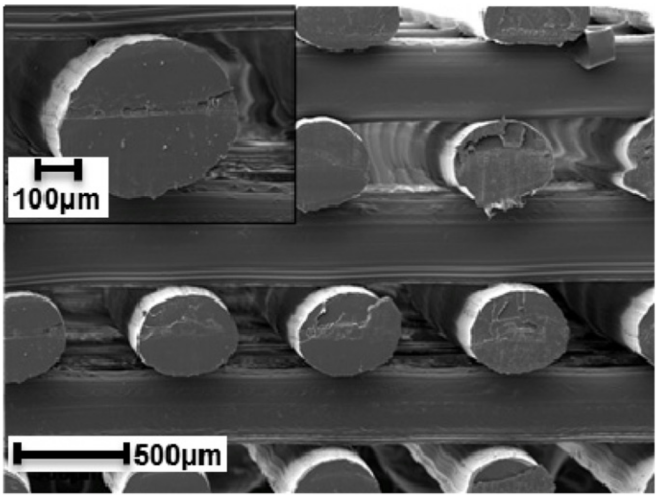



**Figure 7.** SEM image showing the internal architecture, interconnectivity and porosity of PED fabricated scaffolds, scale bar: 100 $\mu$m (top), 500 $\mu$m (bottom).

### *3.2. Biological characterizations*

*3.2.1. Cytotoxicity analysis.* MarkerGene™Live:Dead/Cytoxicity Assay Kit was used to analyze the cytotoxicity of the scaffolds. Manufacturer's protocols were followed to create the working live:dead solution from the propidium iodide (PI) solution and the carboxyfluorescein di-acetate (CFDA) solution. The carboxyfluorescein dye is retained within live cells, producing a green fluorescence, while cells with damaged membranes allow the entrance of PI, which undergoes a fluorescence enhancement upon binding to nucleic acids promoting a red fluorescence in dead cells. Both quantitative and qualitative data were collected on all scaffolds on days 7, 14 and 21 after cells were seeded onto the scaffold.

Figure 8 illustrates live (green) and dead (red) cells within the scaffolds: images A1, A2 and A3 are the fluorescence images of scaffold 1 on days 7, 14 and 21, respectively; images B1, B2 and B3 are the fluorescence images of scaffold 2 on days 7, 14 and 21, respectively; and images C1, C2 and C3 are the fluorescence images of scaffold 3 on days 7, 14 and 21, respectively. As seen in figure 8, live cells greatly outnumber the dead cells on the top surface of the scaffold. Fluorescence images presented in figure 8 provide sufficient evidence that the fabrication process does not change the biocompatibility of the polymer.

Quantitative data were also collected using a microplate reader (GENios, TECAN, North Caroline, USA) with the manufacturer's recommended excitation and emission wavelengths. Figure 9 shows a graph of the data that was collected from the microplate reader. The increasing trend of live cells from day 7 to day 21 for scaffolds 1, 2 and 3 in

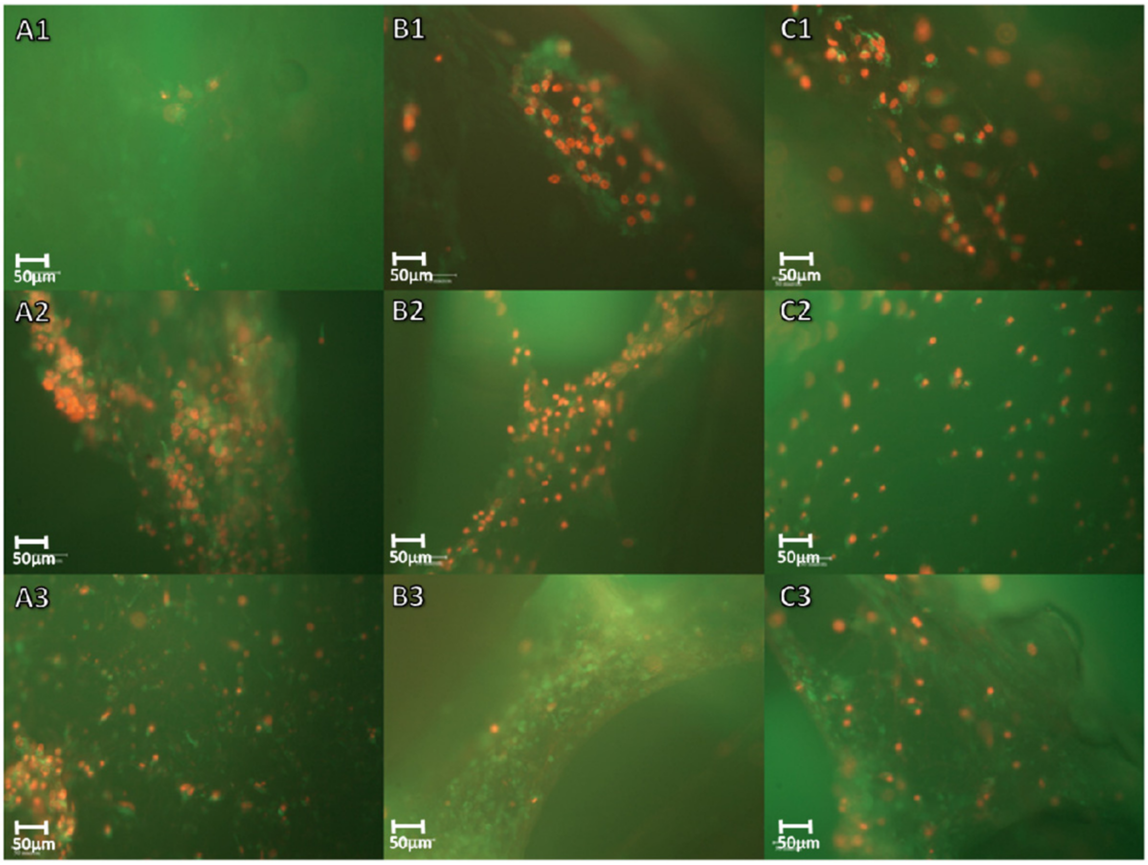



**Figure 8.** A1, A2 and A3 are the fluorescence images of scaffold 1 on days 7, 14 and 21, respectively. B1, B2 and B3 are the fluorescence images of scaffold 2 on days 7, 14 and 21, respectively. C1, C2 and C3 are the fluorescence images of scaffold 3 on days 7, 14 and 21, respectively, magnification: 20×, scale bar 50 $\mu$m.

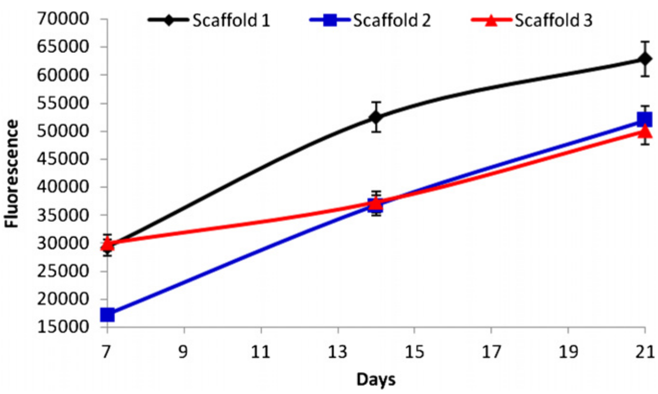



**Figure 9.** Live:dead study of live cells on scaffold 1, 2 and 3 in terms of fluorescence.

figure 9 provides additional evidence of the biocompatibility of the scaffolds.

*3.2.2. Cell–scaffold interaction.* The direct interactions between cells play an important role in the development and function of multicellular organisms [44]. It is important that cells maintain this interaction within scaffolds; without the cell–cell interaction within the scaffolds, cells cannot proliferate and differentiate into mature cells that are essential for functional tissues. Harvested 7F2 cells were seeding onto scaffold 1, 2 and 3 for a period of 21 days. Several biological characterizations were conducted on various checkpoints within those 21 days.

A fluorometric indicator (Alamar Blue, Serotec) of cell metabolic activity was utilized to determine cell proliferation. A cell-laden scaffold was removed from the culture plates, washed twice, placed into a new culture plate where 1 mL of fresh media were added. 10% alamar blue was added to the culture plate and incubated for 4 h. The resulting 1 ml of solution was removed from the culture plate and placed into the microplate reader whose excitation and emission wavelengths were 535 and 590 nm, respectively. This study was conducted on days 3, 5, 7, 11, 14 and 21. A cell number was obtained through a calibration curve determined by correlating a known cell number with the fluorescent intensity of the solution (figure 10).

As seen by the data plotted in figure 11, cells on scaffolds 1, 2 and 3 proliferated over the 21 day period. Cells on scaffold 1 showed the highest level of proliferation; this may be due to the high surface area (in comparison to scaffolds 2 and 3) on the scaffold. Scaffold 3 shows the longest active proliferation; this may be due to the enormous pore size (in comparison to scaffolds 1 and 2) on the scaffold; the large pores provided more 'empty space' for cells to grow. The size of the pores on the scaffolds shows a direct correlation as to the number of cells that are attached on the scaffold during the

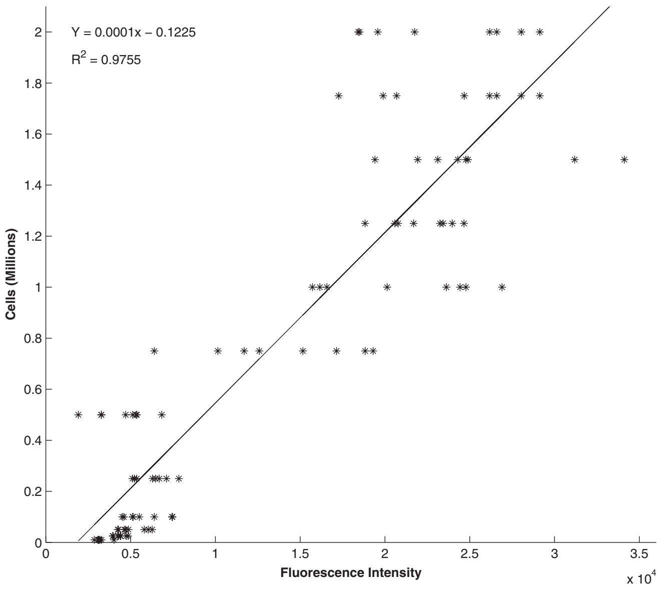



**Figure 10.** Calibration curve of cell number.

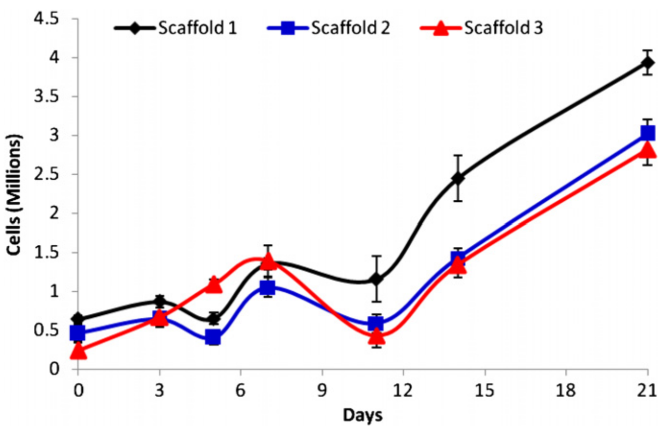



**Figure 11.** Proliferation of scaffolds 1, 2 and 3.

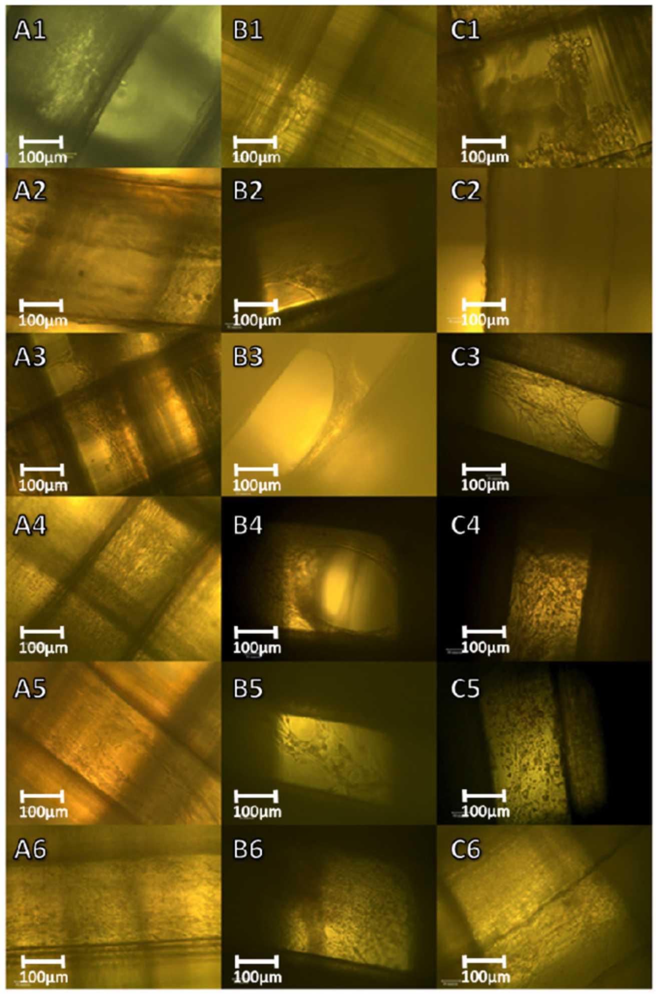



**Figure 12.** Snapshot of cells within scaffolds 1, 2 and 3. A1–A6 show cells within scaffold 1 on days 0, 3, 5, 7, 14 and 21, respectively. B1–B6 show cells within scaffold 2 on days 0, 3, 5, 7, 14 and 21, respectively. C1–C6 show cells within scaffold 3 on days 0, 3, 5, 7, 14 and 21, respectively, magnification: 20×, scale bar 100 $\mu$m.

initial seeding process; scaffold 1 retained the greatest amount of cells (60%) followed by scaffold 2 (43%) and scaffold 3 (23%). Scaffolds 1 and 2 showed similar proliferation trends where there would be a decrease of cells on days 5 and 11, and on day 11 all scaffolds showed a proliferation trend which means that there is a decrease in the cell number. The decrease in cell numbers may be due to the inability of alamar blue to react with the cells trapped within the mineralized matrix; it may be due to the lack of nutrients getting to the cells deep within the scaffold, or it may be due to pores being closed up by proliferated cells and preventing nutrients from getting to other cells on the scaffolds. The data collected also show a great increase of cells after there is a decline in the cell number; this may be due to the extracellular matrices (ECM) created by cells on the scaffold. At the end of the 21 day study, all scaffolds demonstrated an increased proliferation trend from day 14 to day 21.

Figure 12 provides a snapshot of cells on scaffolds 1, 2 and 3 on images labeled A, B and C, respectively, on days 0, 3, 5, 7, 14 and 21 on images labeled X1, X2, X3, X5 and X6, respectively, where 'X' represents A, B or C which correlates with a specific scaffold. As seen on the snapshots, cells are continuously growing and closing the pores within the scaffolds. The snapshots also support the trend line of each scaffold's proliferation study.

*3.2.3. Alkaline phosphatase activity.* Scaffolds were removed from culture plates and washed twice with 1× PBS. The washed scaffolds were then submerged into 1 mL of 1% Triton ×100 solution (Fisher) for cell lysis. Scaffolds were incubated for 1 h and then centrifuged for 10 min at 1000 RPM. Alkaline phosphatase (ALP) activity was calculated by the p-nitro phenyl phosphate (p-NPP) method. An ALP ready solution (Millipore) was utilized in these studies. 200 $\mu$L of the ALP solution was added to 700 $\mu$L of supernatant and incubated for 45 min. The absorbance of this mixture was taken at 405 nm every minute for 60 min using the microplate reader. The absorbance collected was converted

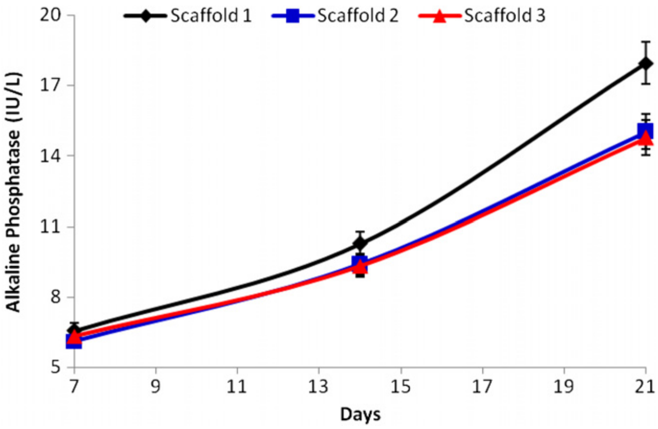



**Figure 13.** Alkaline phosphatase of scaffolds 1, 2 and 3 on days 7, 14 and 21.

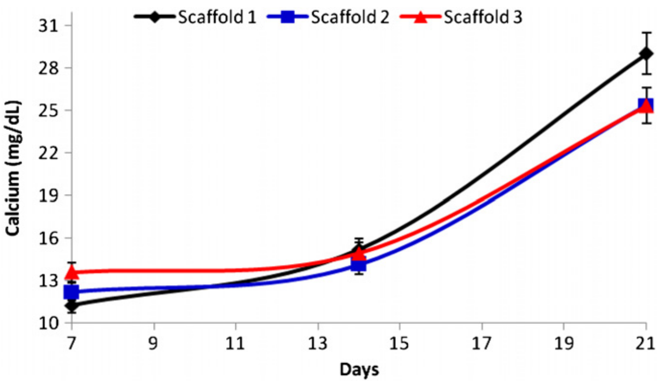



**Figure 14.** Calcium production of scaffolds 1, 2 and 3 on days 7, 14 and 21.

to international units per liter (IU/L) using the conversion calculations provided by the manufacturer. This study was conducted at time points 7, 14 and 21 days for scaffolds 1, 2 and 3, respectively.

As seen in figure 13, there is an upregulation of alkaline phosphatase from day 7 to day 21 on scaffolds 1, 2 and 3. Scaffolds 2 and 3 have just about the same amount of alkaline phosphatase throughout the study, while scaffold 1 has a slightly greater amount of alkaline phosphatase.

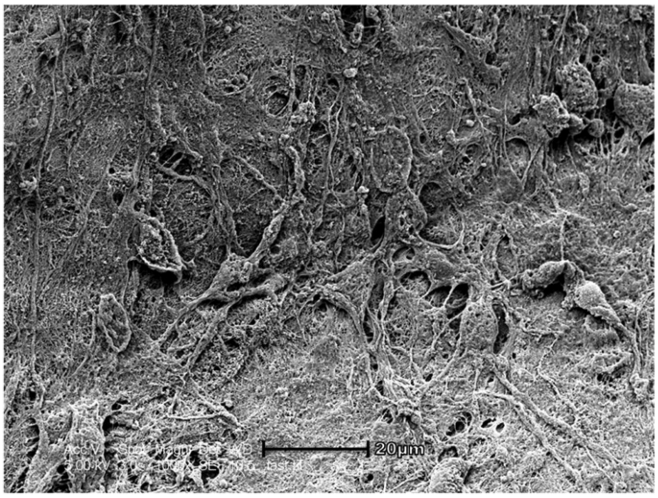



**Figure 16.** 1000× magnification of the cell morphology within the scaffolds, scale bar: 20 $\mu$m.

*3.2.4. Calcium production.* Calcium in the cell-laden scaffolds 1, 2 and 3 was characterized on days 7, 14 and 21 with a calcium test kit (Stanbio Laboratories, TX, USA). Scaffolds were washed twice with 1× PBS. The washed scaffolds were submerged in a cell lysis solution (0.5 mL Triton ×100 with 0.5 mL trichloroacetic acid) and incubated for 1 h and then centrifuged for 10 min at 1000 rpm. Characterizations were conducted according to the manufacturer's protocol; 10 $\mu$L of supernatant was added to the calcium test working solution (500 $\mu$L 2-amino, 2-methyl, 1-propanol (base reagent) with 500 $\mu$L o-cresolphthalin (color reagent)). Absorbance was taken at 550 nm with a microplate reader. A standard absorbance was also taken for the conversion of absorbance to mg $dL^{-1}$.

As seen in figure 14, there is an upregulation of calcium from day 7 to day 21 on scaffolds 1, 2 and 3. Even though scaffold 1 had the lowest amount of calcium on day 7, at the end of the study, scaffold 1 had the greatest amount of calcium, while scaffolds 2 and 3 have just about the same amount of calcium throughout the study.

*3.2.5. Morphological study.* SEM images were taken of the cell-scaffold on day 21 to illustrate the cell morphology within

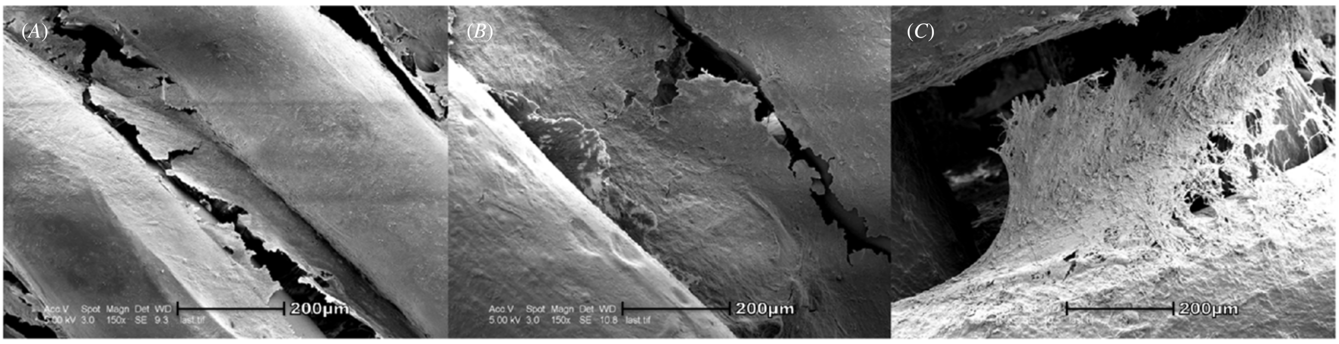



**Figure 15.** Cell morphology on day 21 of (*A*) scaffold 1, (*B*) scaffold 2, (*C*) scaffold 3, scale bar: 200 $\mu$m, magnification: 150×.

scaffolds 1, 2 and 3. On day 21, scaffolds 1, 2 and 3 were removed from the cell culture medium and washed twice with 1× PBS. Each scaffold was fixed with 4% glutaraldehyde (Sigma Aldrich, USA) for 2 h. Each scaffold was then subjected to dehydration by submerging each scaffold for 10 min through a series of diluted ethanol (20%, 50%, 70%, 90%, 95% and 100%). Scaffolds were then refrigerated at 4 °C for 24 h. Prior to SEM characterization, each scaffold was coated with approximately 10–20 nm of platinum.

As illustrated in figure 15, on day 21, each scaffold has a significant amount of cells on the filament and within the pores. In addition, as the pore increases in size, the spread of cells within the pores becomes less dense. Figure 16 shows a 1000× magnification of the cells within the scaffolds. Each scaffold illustrates the same cell morphology seen in figure 16.

## 4. Conclusions

This study investigated the PED capabilities to fabricate the three-dimensional scaffold with the use of the AC device. As illustrated in this paper and previous papers [1, 2], the PED device has proven successful in fabricating three-dimensional scaffolds. However, scaffolds previously fabricated with the PED utilized polymers with a low melting point. To utilize the PED's full potential, the AC device was integrated. This allowed the PED to utilize a wider range of biopolymers with higher melting points. To demonstrate that the integration was beneficial and successful, a high melting point biopolymer (proprietary) was chosen for investigation. The SEM characterization illustrated the integrated device ability to fabricate three-dimensional scaffolds of different porosity. In addition, a series of biological characterizations demonstrated the scaffold's ability to stimulate cell attachment, proliferation and differentiation. In conclusion, the PED with integrated AC device is capable of fabricating biological scaffolds for the applications of tissue engineering and regenerative medicine.

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